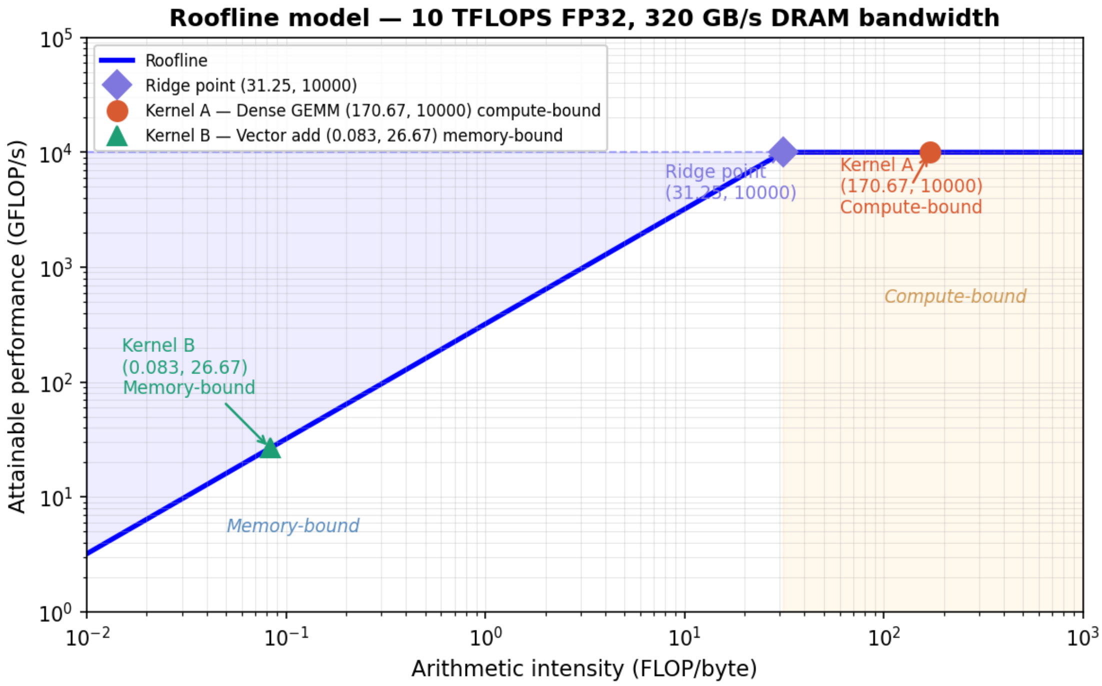
1.



2. Kernel A: Dense GEMM

- Task: Multiplying two 1024×1024 FP32 matrices.
- Total FLOPs: 2 x N^3 = 2 x 1024^3 = 2,147,483,648 FLOPs
- Total Bytes: Loading two matrices and storing one (3 x 1024^2 x 4 bytes) = 12,582,912 bytes
- Arithmetic Intensity (AI): 2,147,483,648 / 12,582,912 = 170.67 FLOP/byte
- Classification: Compute-Bound. Since 170.67 > 31.25, the kernel hits the flat compute ceiling.

3. Kernel B: Vector Addition

- Task: C = A + B for FP32 vectors of length 4,194,304.
- Total FLOPs: 1 addition per element = 4,194,304 FLOPs
- Total Bytes: Loading two vectors and storing one (3 x 4,194,304 x 4 bytes) = 50,331,648 bytes
- Arithmetic Intensity (AI): 4,194,304 / 50,331,648 = 0.0833 \FLOP/byte
- Classification: Memory-Bound. Since 0.0833 < 31.25, the kernel is limited by the slanted bandwidth roof.

4. Architectural Recommendations

- For Kernel A: To improve performance, I recommend increasing peak compute throughput. This can be achieved by adding more FP32 ALUs or implementing specialized hardware like Tensor Cores to handle matrix math more efficiently.
- For Kernel B: To improve performance, I recommend increasing memory bandwidth. This can be achieved by utilizing High Bandwidth Memory (HBM) or adding larger on-chip caches to reduce the frequency of DRAM accesses.